SI 315

Models of Social Information Theory

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6 Network Searching

6.1 Hubs and Authorities

- Used to identify important nodes in a network
- Hubs are nodes that point to many authorities
- Authorities are nodes that are pointed to by many hubs

6.1.1 Measuring Rankings

- Authority score: sum of scores of hubs pointing to the authority
- Hub score: sum of scores of authorities pointed to by the hub
- HITS: iteratively update authority and hub scores

6.1.2 Example Using Adjacency Matrix

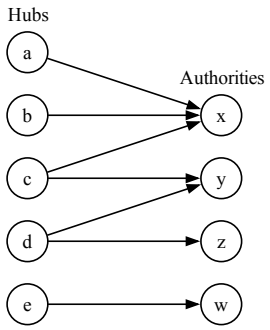


Figure 6.1: Bipartite network of hubs pointing to authorities.

- Node order: hubs (a, b, c, d, e); authorities (x, y, z, w)
- Adjacency matrix A: rows = hubs; columns = authorities

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Initialize hub scores: $\mathbf{h}^{(0)} = (1, 1, 1, 1, 1)$
- Each iteration: $\mathbf{a}^{(t)} = A^{\mathsf{T}}\mathbf{h}^{(t-1)}$; $\mathbf{h}^{(t)} = A\mathbf{a}^{(t)}$
- Integer updates below: no intermediate normalization; reported scores normalized to unit L_2 length

- Iteration 1:
$$\mathbf{a}^{(1)} = A^{\mathsf{T}}\mathbf{h}^{(0)} = (3, 2, 1, 1), \quad \widehat{\mathbf{a}}^{(1)} \approx (0.775, 0.516, 0.258, 0.258)$$
$$\mathbf{h}^{(1)} = A\mathbf{a}^{(1)} = (3, 3, 5, 3, 1), \quad \widehat{\mathbf{h}}^{(1)} \approx (0.412, 0.412, 0.687, 0.412, 0.137)$$
- Iteration 2:
$$\mathbf{a}^{(2)} = A^{\mathsf{T}}\mathbf{h}^{(1)} = (11, 8, 3, 1), \quad \widehat{\mathbf{a}}^{(2)} \approx (0.788, 0.573, 0.215, 0.072)$$
$$\mathbf{h}^{(2)} = A\mathbf{a}^{(2)} = (11, 11, 19, 11, 1), \quad \widehat{\mathbf{h}}^{(2)} \approx (0.409, 0.409, 0.706, 0.409, 0.037)$$
- Iteration 3:
$$\mathbf{a}^{(3)} = A^{\mathsf{T}}\mathbf{h}^{(2)} = (41, 30, 11, 1), \quad \widehat{\mathbf{a}}^{(3)} \approx (0.789, 0.577, 0.212, 0.019)$$
$$\mathbf{h}^{(3)} = A\mathbf{a}^{(3)} = (41, 41, 71, 41, 1), \quad \widehat{\mathbf{h}}^{(3)} \approx (0.408, 0.408, 0.707, 0.408, 0.010)$$
- Iteration 3 ranking: c strongest hub; x strongest authority
- Ranking trend: c and x dominant; e and w increasingly weak

6.2 PageRank

- Method for ranking nodes in a network based on the structure of incoming links
- Counts the number and quality of links to a page to determine its importance

1. Assign equal scores to all nodes: $\frac{1}{n}$
2. Update scores based on the scores of incoming links, typically using the formula: $\mathrm{PR}(i) = \sum_{j \in M(i)} \frac{\mathrm{PR}(j)}{L(j)}$ where $M(i)$ is the set of nodes linking to i and $L(j)$ is the number of outbound links from node j